Pre-Job Inspection

Which fields are required on every visit, and which ones only some jobs unlock.

Version 1 · For build · Otopair internal

The one rule everything below comes from

A field is required on every visit **only** if a mechanic can answer it honestly on a car sitting in the bay — wheels on, no lift needed, no tool beyond a tread gauge and a tire gauge — in about six minutes total.

If a field fails that test and we mark it required anyway, we do not get data. We get a mechanic clicking green to clear the form. A fake green is worse than a blank, because we cannot tell it apart from a real one.

The five tiers at a glance

Tier	When it is asked	Example
Tier 1	Every visit. Every vehicle. No exceptions.	Tread depth, oil level, lights
Tier 2	Only when a wheel is already coming off for the job.	Pad thickness, rotor thickness
Tier 3	Only when the car is already on a lift.	CV boots, exhaust, leaks
Tier 4	Only when the job opens that system.	Oil viscosity, coolant strength
Tier 5	First visit, then again only when something changed.	Tire brand, DOT date

The plain-English version of the tier split

On a brake job we do **not** ask what oil viscosity is in the car. We **do** ask whether the oil level is fine. Level is free on every car with the hood up. Viscosity is not, and it does not matter unless we are opening that system.

That single example is the whole document.

Tier 1 — Always required

Sixteen items. Every vehicle, every service, no exceptions. Nothing here needs a lift, a wheel off, or a special tool. This is the entire mandatory set — if a field is not on this list, it is not required.

Field	What the mechanic actually does	Answer type
Odometer	Reads the dash. Saved against this inspection, not just the booking.	Number (miles)
Warning lights	Notes which lights are lit. He already drove it into the bay.	Checklist, or "none"
Tread depth × 4	Gauge into the tread on each tire.	Number (/32")
Tire pressure × 4	Gauge on each valve stem.	Number (psi)
Tire condition × 4	Looks at the tread face and sidewall — uneven wear, cracks, damage.	Good / Monitor / Attention
Tires match?	Looks at all four sidewalls. Brand, model and size.	All match / One odd / Mixed on an axle
Engine oil level	Pulls the dipstick. Level only — not condition, not viscosity.	Full / Low / Empty
Coolant level	Looks at the reservoir. Does not open anything.	Full / Low / Empty
Brake fluid level	Looks at the reservoir against the MIN / MAX marks.	Full / Mid / Low
Washer fluid level	Looks at the reservoir.	Full / Low / Empty
Brake state × 4	Looks through the wheel at each corner. State only, never a number.	Good / Monitor / Attention / Not visible
Exterior lights	Turns them on and walks around: head, tail, brake, turn, hazards.	All pass, or lists which are out
Windshield	Looks for chips and cracks.	Good / Monitor / Attention
Wiper blades	Looks at the rubber and feels the edge.	Good / Monitor / Attention
Horn	Presses it.	Works / Does not
Battery terminals	Looks under the hood for corrosion. Visual only.	Clean / Corroded

Do not skip this one — brake fluid level is a free brake-wear signal

As brake pads wear down, the caliper pistons push further out and brake fluid moves out of the reservoir to fill the space behind them. So the reservoir level slowly drops as the pads wear.

That means a falling brake fluid level, with no leak anywhere, tells us the pads are wearing — **without anyone removing a wheel.** It takes ten seconds and the hood is already up.

To use it we need the level stored as a **reading we can compare to last visit**, not just a green dot. Store the level, compare against the previous inspection, and flag the drop.

Tier 2 — Only when a wheel is off

Unlocks when: the job already requires a wheel to come off — brake service, tire mounting, suspension work, or a state inspection where the mechanic pulls a wheel.

These are never required on an oil change. Pad and rotor thickness cannot be honestly measured with the wheel on: the inner pad wears differently from the outer and you cannot see it through the spokes, and the rotor needs a micrometer against the face. Even on a lift, the wheel is in the way.

Field	What the mechanic does	Answer type
Pad thickness	**Inner and outer separately**, each corner. Two numbers per corner, not one.	Number (mm) × 2
Rotor thickness	Micrometer against the rotor face. Records which tool he used.	Number (mm) + tool
Rotor stamp	Types in exactly what is stamped on the rotor, and photographs it.	Text + photo
Rotor condition	Tags what he sees.	scored / pitted / rusted / warped / grooved
Caliper slides + boots	Checks the slide pins move and the boots are intact.	Good / Monitor / Attention
Brake hose	Looks for cracking, bulging, weeping.	Good / Monitor / Attention

> **Build the rotor stamp as text and a photo. Never as a "minimum thickness" number field.**
> The number stamped on a rotor does not mean the same thing on every car.
>
> European cars stamp **MIN TH**, and that is the true discard point — past it, the rotor is scrap.
> Many US cars stamp the **minimum after machining** instead, which is a **larger** number.
>
> If we build one field called "minimum thickness" and let the mechanic type a number into it, we will mix those two meanings together and be wrong by roughly half a millimetre on domestic cars — in the unsafe direction. So: capture the stamp **exactly as written**, plus a photo. The backend decides which kind it is.

Tier 3 — Only when the car is on a lift

The current Underbody section is required on every visit. It should not be. None of it can be answered properly on a car sitting on its wheels. Split it in two:

Group	Unlocks when	Fields
3A · Visible from underneath	The car is on a lift, wheels still loaded.	Fluid leaks and drips · torn CV boot · leaking strut · exhaust leak or broken hanger · undercarriage damage
3B · Needs the wheels hanging	The job already has the wheels off the ground.	Steering linkage play · ball joint play · wheel bearing play

Tier 4 — Only when the job opens that system

These are the "condition and specification" checks, as opposed to the "level" checks in Tier 1. They cost real time and they only matter when we are already in that system.

Field	Only ask when the job is	Note
Oil type / viscosity + condition	Oil service	Level stays in Tier 1. This is the rest of it.
Coolant strength / test	Cooling system work	Needs a refractometer or test strip.
Brake fluid moisture test	Brake service	Needs test strips.
Transmission fluid	Transmission service	**Must allow N/A.** Many modern cars are sealed with no dipstick.
Power steering fluid	Steering work	**Must allow N/A.** Many cars are electric — no fluid at all.
Battery load test (CCA)	No-start, electrical complaint, or seasonal check	Terminals stay visual in Tier 1.
Engine air filter	Filter or tune service	On some cars this is a real job, not a 60-second check.
Cabin air filter	Filter service	Often behind the glovebox. Not a per-visit item.

> **Transmission and power steering are currently required with no way to say "this car does not have one"**
>
> A sealed transmission has no dipstick. An electric power steering rack has no fluid. On those cars the mechanic physically cannot answer, so he will click green and move on — and we will store a green that means nothing. Both fields need a real **N/A** option before they ship.

Tier 5 — Captured on the first visit, then only when something changes

Tire brand, model, size, DOT date code, run-flat yes/no, and pad brand/type. We do not ask the mechanic to type these on every visit — that is exactly the kind of friction that gets faked. We ask for them when something actually changed.

Two triggers, either one fires the capture:

Trigger	What happens
Mechanic answers "one odd" or "mixed on an axle" on the Tier 1 match question	Form expands and asks for brand, model and size on the odd corner only.
Tread depth on any corner reads higher than the last visit	That corner got a new tire. System asks for brand, model, size and DOT date on that corner only.

The tread-depth trigger costs us nothing and cannot be forgotten

Tread depth only ever goes down in the real world. So if a corner reads 4/32" in March and 9/32" in July, a tire was replaced — the system knows before anyone tells it.

We already collect tread depth on all four corners every visit in Tier 1. Comparing it to the previous reading is one subtraction. No extra work for the mechanic, and the tire record can never go quietly stale.

Four build rules — none of the above works without these

Rule	Why
1. Nothing pre-fills. Every field starts empty.	The current form seeds every field with a plausible value — tread 6, pressure 33, pad 5, rotor 24.1, and every dot pre-set. A mechanic can tap through seven zones without touching a single field and submit a complete, believable, entirely made-up inspection. This is the most important fix in the document.
2. Every field needs "Not inspected" and "N/A", and they are different things.	"Not inspected" means nobody looked. "N/A" means the car does not have one. Neither is green. Right now the honest answer has nowhere to go, so it becomes a green.
3. Green never clears a flag.	A green visual check can lower urgency. It can never close an open item. Only a real measurement or a completed service closes one. This has to be enforced in state, not left as something we all remember.
4. Every number stores how it was taken.	Micrometer / gauge / visual / not measured. Without this we cannot tell a real measurement from a guess, and we can never promote a reading to trusted data.

What changes in the form as it is built today

Change	Detail
Pad + rotor thickness	Currently required on all four corners. Change to Tier 2 — wheel-off jobs only.
Add brake state per corner	New tri-state with a fourth option: Good / Monitor / Attention / **Not visible**.
Underbody section	Currently required. Change to Tier 3 and split into 3A (visible) and 3B (wheels hanging).
Transmission + power steering	Add N/A.
New Tier 1 fields to add	Odometer on the record · warning lights · washer fluid · horn · "tires match?"
Brake fluid	Store as a comparable level reading, not a green/yellow/red dot.
Remove all default values	Every field empty on load.
"Mark zone complete"	Block it until the required fields in that zone have actually been touched.
Tire brand / model	Move off "front-left only, always" and onto the Tier 5 change triggers, any corner.

Questions already asked, and the answers

So we do not go around this again.

Do brake pads not have a wear indicator we can just look at?

Both things are true and they are different claims. A **visual** wear groove is inconsistent and often not visible through the wheel — that part is right. But a wear indicator in general is close to universal on modern cars: either a metal squealer tab that scrapes the rotor and squeals, or an electronic sensor that lights the dash.

The reason it still does not solve our problem is **coverage**. Usually only one pad on an axle carries the tab. Many cars have no rear squealer at all. Electronic sensors are normally fitted to one corner per axle. So a car that is not squealing is not a car with good pads — it is a car whose one sensored corner has not triggered yet.

That is exactly why the Tier 1 corner check is a **state** and never a clear. It can raise a flag. It cannot close one.

Can we just get real measurements once a year at the state inspection?

Two problems.

Shops in practice. Inspections in NY get run around emissions, tint and the front plate. Brake work at inspection gets skipped, and the regulation has an "approved brake tester" exemption that lets a shop legally not pull the wheel. We should not build a data pipeline that assumes shops do the optional part.

One reading a year is not a wear rate. A single thickness number is a snapshot. To project when something is due we need two readings with mileage attached. At one per year, the first useful projection lands in month 24. That is an archive, not a timing engine.

The fix is not a bigger annual check. It is many small honest checks (Tier 1) plus real measurements whenever a wheel happens to be off anyway (Tier 2).

Why is engine oil level required on a brake job?

Because it is free. The hood is open, the dipstick takes twenty seconds, and a low oil level matters to the customer no matter what they came in for. What we do **not** ask on a brake job is oil viscosity or oil condition — those cost real time and only matter if we are draining it.

Level is Tier 1. Condition and specification are Tier 4. That distinction runs through the whole document.

Why not just require everything and let the mechanic skip what he cannot do?

Because there is currently no way for him to say "I could not check this." His only options are green, yellow, red — so an impossible check becomes a green. Once that happens we cannot tell a real green from a shrug, and the whole dataset loses its meaning. Build rule 2 fixes this, and the tiers make sure we are not asking the question in the first place.

Not in this version: escalation rules based on drivetrain (treating a tire mismatch differently on an all-wheel-drive car than a front-wheel-drive one). Parked deliberately — it is a separate pass and it does not change any field in this document.